

From Frozen Encoders to Multi-Agent LLM Pipelines: An Iterative Journey through Person–Place Relation Extraction in Multilingual Historical Texts

Team Hansel&Gretel at CLEF HIPE-2026

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Abstract

We present the system description and experimental analysis of Team Hansel&Gretel for the CLEF HIPE-2026 Shared Task on Person–Place Relation Extraction from Historical Multilingual Documents. The task requires classifying, for each person–location pair in a historical newspaper article (German, French, or English, spanning the 19th–20th centuries), two temporal relations: *at* (person was ever at the location, 3-class) and *isAt* (person was there within approximately one month of publication, binary). Evaluation uses Macro Recall, equally weighting all label classes. We explore nine progressively sophisticated strategies, from a frozen XLM-RoBERTa encoder with classical ML classifiers (internal dev Global Macro Recall = 0.6018) to a Qwen2.5-3B-Instruct Generative Supervised Fine-Tuning pipeline with Wikidata knowledge injection (dev = 0.7289) and a three-agent Historian–Geographer–Arbiter multi-agent architecture (dev = 0.7280). On the official competition test set, our best run (Run 3: GPT-OSS 120B with Chain-of-Thought and Wikidata injection) achieves an impresso score of **0.6221**, ranking 13th out of 46 ranked entries (45 system runs + random baseline). This paper documents the design decisions, failure modes, and key insights behind each strategy, including an honest analysis of the 17-point dev-to-test gap, providing a comprehensive account of iterative NLP system development under data scarcity, class imbalance, and multilingual historical OCR noise.

Keywords

Person–Place Relation Extraction, Historical NLP, Multilingual Transformers, XLM-RoBERTa, LoRA, Generative Classification, Multi-Agent LLM, Wikidata, HIPE-2026, CLEF

1. Introduction

The digitisation of historical newspapers has created vast corpora of 19th and 20th century text encoding biographical trajectories and cultural history. Automatically recovering *where people were and when* from these documents enables digital humanities scholars to reconstruct spatio-temporal life trajectories at scale. This is the central challenge of **CLEF HIPE-2026** [1, 2]: the Shared Task on Person–Place Relation Extraction from multilingual historical documents, the third edition in the HIPE series following HIPE-2020 [3] and HIPE-2022 [4].

Unlike standard named entity recognition, HIPE-2026 requires *temporal reasoning*: a system must determine whether a text provides evidence for a *historical* presence (*at*) or a *contemporary* one (*isAt*, bounded to ≈ 1 month before publication). This distinction, combined with multilingual OCR-noisy text, severe class imbalance, and a small labelled dataset, makes the task deceptively difficult.

Team Hansel&Gretel approached this as an iterative research problem, developing **nine distinct strategies** that form a natural progression:

1. **S1**: Frozen XLM-RoBERTa + Classical ML classifiers
2. **S1.5**: Multi-Task Learning (MTL) + Optuna Hyperparameter Optimisation
3. **S2**: Full End-to-End XLM-RoBERTa Fine-Tuning
4. **S2.5**: PEFT/LoRA on XLM-RoBERTa

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5. **S3**: Monolingual LoRA Routing (language-partitioned adapters)
6. **S4/S4.5**: Qwen2.5-0.5B MTL with LoRA (+ soft label variant)
7. **S5**: Qwen2.5-3B-Instruct Generative SFT (*best dev score*)
8. **S6/S7**: Three-specialist multi-agent pipeline (Historian/Geographer/Arbiter)
9. **S8/S9**: Chain-of-Thought few-shot via IBM RITS large LLMs + Wikidata injection (*best official test score*)

The paper is structured as follows: Section 2 defines the task formally. Section 3 analyses the dataset. Section 4 describes the organiser baseline. Sections 5–9 present each strategy group. Section 10 provides the cross-strategy comparison. Section 11 reports official test results. Section 12 presents error analysis. Sections 13–15 discuss limitations, future work, and conclusions.

2. Task Description

2.1. Formal Definition

HIPE-2026 provides historical newspaper documents as input. Each document contains full article text with metadata (title, publication date, language) and a list of sampled (person, location) entity pairs. For each pair, systems classify two relations:

$$f(\text{context}, (\text{person}, \text{location})) \mapsto \left(\underbrace{\text{at}}_{\in \{\text{TRUE}, \text{PROBABLE}, \text{FALSE}\}}, \underbrace{\text{isAt}}_{\in \{\text{TRUE}, \text{FALSE}\}} \right) \quad (1)$$

The **at** relation is ternary: TRUE if the text explicitly places the person at the location at any past time; PROBABLE if inferable from implicit cues; FALSE otherwise. The **isAt** relation is binary: TRUE only within ~ 1 month of publication. A hard *transitivity constraint* applies: $\text{isAt} = \text{TRUE} \implies \text{at} \in \{\text{TRUE}, \text{PROBABLE}\}$.

2.2. Evaluation Metric

The primary metric is **Macro Recall** (also called balanced accuracy [5]), computed independently per sub-task:

$$\text{MacroRecall} = \frac{1}{|L|} \sum_{\ell \in L} \frac{\#\text{correctly predicted with label } \ell}{\#\text{examples with gold label } \ell} \quad (2)$$

The **impresso score** averages both sub-task macro recalls. Three evaluation profiles are assessed: **Accuracy** (Macro Recall on impresso newspaper Test Set), **Efficiency** (composite rank: accuracy + parameter count + model size), and **Generalisation** (Macro Recall on surprise literary Test Set, at only).

3. Dataset Analysis

3.1. Data Sources and Splits

The dataset derives from HIPE-2022 [4] entity-annotated corpora, re-annotated for person–place relation extraction. It covers four historical newspaper collections (*hipe2020*, *sonar*, *newseye*, *letemps*) in German, French, and English spanning 1780–1950, distributed through the Impresso Media Monitoring of the Past project [6].

In addition, a **silver-labelled sandbox** corpus (~ 461 docs, 6,170 pairs) of automatically annotated documents provides auxiliary training signal. All our internal evaluations use a subset of the dev set (254 pairs, 21 documents) corresponding to the early-release dev data available during the main experimental phase; the full dev split (159 docs, 2,137 pairs) became available only near the submission deadline. The official hidden test set was released separately and scored by the organisers.

Table 1

Gold dataset statistics by language split (v1.0).

Lang.	Train Docs	Train Pairs	Dev Docs	Dev Pairs
German (DE)	97	1,345	32	432
French (FR)	328	4,521	109	1,557
English (EN)	56	501	18	148
Total	481	6,367	159	2,137

3.2. Exploratory Data Analysis (EDA)

We performed detailed EDA on the gold training split available at experiment time (104 documents, 1,251 pairs):

Table 2

Gold training split EDA statistics (104 documents, early experiment phase).

Metric	Value
Total candidate pairs	1,251
Language split	34 DE / 35 EN / 35 FR
Unique persons	496
Unique locations	469
Average document length	3,465 chars
Maximum document length	11,738 chars
at: TRUE / PROBABLE / FALSE	441 / 120 / 690
isAt: TRUE / FALSE	279 / 972

Three challenges shaped all modelling decisions:

1. **Context Length:** Max $\sim 12,000$ chars ($\approx 3,000$ tokens) exceeds standard 512-token encoder limits, motivating generative LLMs.
2. **Class Imbalance:** isAt=FALSE at 77.7%; PROBABLE at 9.6% of at labels. Macro Recall penalises naive majority-class predictors heavily.
3. **Multilingual by Design:** Equal DE/EN/FR split — monolingual models immediately lose cross-lingual transfer.

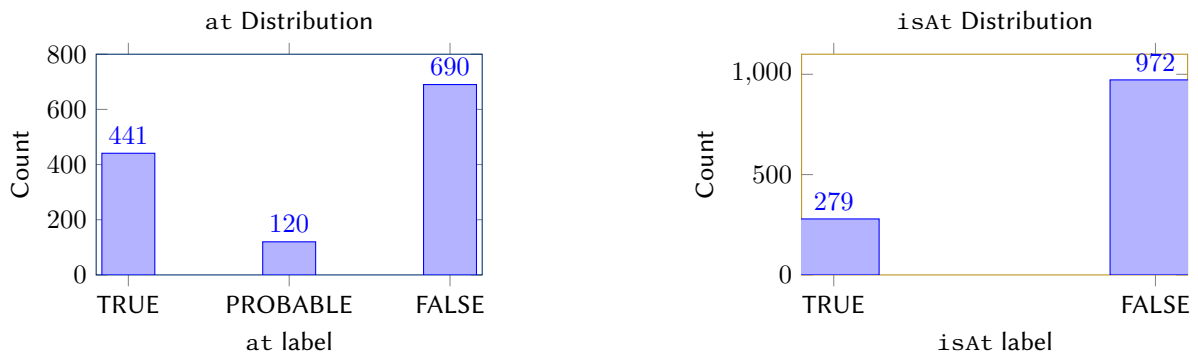


Figure 1: Label distributions in the gold training set. Severe imbalance in isAt (78% FALSE) drives the choice of Macro Recall as the evaluation metric.

3.3. Why This Dataset Is Particularly Challenging

The HIPE-2026 data presents a unique confluence of difficulties rare in modern NLP benchmarks: (1) **OCR noise** from historical typesetting (e.g., “Mufschén” for “München”); (2) **archaic vocabulary** and

grammar absent from modern pre-training corpora; (3) **temporal reasoning** over implicit signals (verb tense, temporal adverbs, publication dates); (4) **entity ambiguity** (same mention, different individuals); and (5) **data scarcity** (~ 100 gold documents across 3 languages and 3 label classes at experiment time).

4. Organiser Baseline

The HIPE-2026 organisers provided a minimal LLM baseline [7] using `mistralai/Ministral-3-3B-Instruct-2512` (GGUF quantised) via `llama-cpp-python`, zero-shot with greedy decoding. On the official test set, it achieved an impresso score of **0.5818** (rank 17/46 runs). On the internal dev set (254 pairs), the random baseline yielded: `at` MR = 0.3211, `isAt` MR = 0.5273, **Global MR** = 0.4242, establishing the floor.

5. Strategies S1 and S1.5: Frozen Encoder + Classical ML

5.1. S1: XLM-RoBERTa Feature Extraction

Motivation. With only 104 gold documents, fine-tuning 270M parameters risks catastrophic overfitting. S1 uses frozen `xlm-roberta-base` [8] as a feature extractor and trains lightweight classical classifiers on top.

Architecture. Person mentions are wrapped in `<E1>...</E1>` and location mentions in `<E2>...</E2>` marker tokens added to the vocabulary. Contextual hidden states at these marker positions are concatenated into a 1536-dim relation-aware feature vector. A `RandomForestClassifier` (scikit-learn) predicts `at`; a `LGBMClassifier` [9] predicts `isAt`. A post-hoc transitivity rule forces `isAt=FALSE` when `at=FALSE`.

Silver data weighting. Both gold (104 docs) and sandbox silver (617 docs) are used. Gold samples receive $50\times$ **sample weight** so annotations dominate the decision boundary while silver provides historical vocabulary coverage.

Table 3

S1 results on internal dev set. Silver data gave +34.9% gain over gold-only.

Configuration	<code>at</code> MR	<code>isAt</code> MR	Global MR
Gold-only (80/20 split)	0.3725	0.5194	0.4459
Gold + Silver ($50\times$)	0.5200	0.6835	0.6018

5.2. S1.5: Multi-Task Learning + Optuna HPO

Motivation. Two independent classifiers cannot exploit the `at`–`isAt` logical correlation. A shared neural network with two heads learns this jointly.

Architecture. A single PyTorch MTL network on top of frozen XLM-R embeddings: shared dense + GELU + dropout, then two linear heads with combined loss $\mathcal{L} = \alpha \mathcal{L}_{\text{at}}^{\text{WCE}} + (1-\alpha) \mathcal{L}_{\text{isAt}}^{\text{Focal}}$ [10]. Optuna [11] (50 trials, TPE sampler) tunes α , learning rate, dropout, hidden dim, and per-class weights. Full best-trial hyperparameter values are available in our public code repository.¹

6. Strategies S2–S4.5: Discriminative Fine-Tuning Attempts

6.1. S2: Full End-to-End Fine-Tuning (Catastrophic Failure)

Full fine-tuning of `xlm-roberta-base` at `lr` = 2×10^{-5} (10 epochs, batch 8) caused **catastrophic mode collapse**: the model predicted TRUE for every `at` and FALSE for every `isAt` (Global MR = 0.4167,

¹Code: <https://github.com/ikshv4ku/HIPE-2026-Team-Hansel-Gretel>

Table 4

S1.5 vs S1 on internal dev set. MTL jointly modelling the at-isAt correlation provides consistent gain.

Strategy	at MR	isAt MR	Global MR
S1 (RF + LightGBM)	0.5200	0.6835	0.6018
S1.5 (MTL + Optuna)	0.5486	0.7013	0.6249

below S1.5 by -0.2082). Three root causes:

1. **Catastrophic forgetting**: Unfreezing 270M params on only 700 documents destroyed multilingual representations.
2. **Gradient interference**: Single batches mixing archaic German (1798), Victorian English (1860), and 19th-century French created destructively interfering gradient updates.
3. **Head gradient dominance**: Randomly initialised classification heads produced large gradients overwhelming the pre-trained transformer layers.

6.2. S2.5: PEFT/LoRA on XLM-RoBERTa (Partial Recovery)

LoRA [12] (rank=8, $\alpha=16$, 0.11% trainable params) prevented complete mode collapse (Global MR = 0.4541), restoring partial minority-class recall. However, performance remained far below the frozen baseline (S1.5: 0.6249) due to data scarcity: ~ 230 docs per language is insufficient to adapt attention matrices without overfitting.

6.3. S3: Monolingual LoRA Routing (Regression)

Three separate LoRA adapters (DE, EN, FR) with document-language-based routing at inference time eliminates cross-lingual gradient conflicts but **worsened** performance (Global MR = 0.4232). The key insight: by partitioning by language, each adapter trains on only ~ 233 documents. The shared multilingual volume in S2.5 outweighs the harm from cross-lingual interference. **Data scarcity, not language mixing, is the binding constraint.**

6.4. S4 / S4.5: Qwen2.5-0.5B MTL with LoRA

Pivoting to decoder-only architecture: Qwen2.5-0.5B [13] (500M params, 32k token context) with LoRA ($r=16$, $\alpha=32$) and MTL heads. This matched S1.5 performance (Global MR = 0.6264) while handling long documents without truncation. Adding label smoothing ($\epsilon = 0.15$) in S4.5 slightly regressed to 0.6212 — smoothing added noise without compensating signal on the minority PROBABLE class.

Key observation: Discriminative approaches plateau at ~ 0.63 . The small dataset and inherent label ambiguity (especially for PROBABLE) limit classification head performance regardless of backbone.

7. Strategy 5: Generative SFT — Best Dev Performance

7.1. Motivation: The Generative Paradigm Shift

Temporal reasoning (“is the person at the location *right now*?”) requires multi-step inference: locate the event in time, compare it to the publication date, and assess the temporal window. Discriminative classifiers select labels but never reason through *why*. Strategy 5 makes a decisive pivot: the model *generates* structured JSON answers, forcing explicit decision-making.

7.2. Model and Architecture

We use Qwen2.5-3B-Instruct [13] (3.1B params) in a Generative Supervised Fine-Tuning (SFT) regime with LoRA ($r=16$, $\alpha=32$, all linear layers). We initially targeted Gemma-2-2B-IT but pivoted due to Hugging Face gating restrictions — Qwen2.5-3B-Instruct proved superior: non-gated, larger, and with stronger multilingual instruction following.

Each training sample is formatted as a chat-template conversation. The user provides document metadata and text; the assistant generates:

```
{"at": "TRUE", "isAt": "FALSE"}
```

Loss masking: Only the JSON answer tokens contribute to the loss. The model learns the decision format without being distracted by document noise.

7.3. Engineering Challenges

- **Memory:** 3B bf16 model (~ 7 GB) required gradient checkpointing and `expandable_segments=True` to fit on a single 24 GB RTX A5000.
- **Disk:** Root partition exhausted by the 16 GB virtual environment; relocated to `/mnt/combined` via symbolic link.
- **Wikidata hard rule:** At inference, each person’s `pers_wikidata_QID` is queried for death date. If the person died >30 days before publication, `isAt` is forced to `FALSE`. A local JSON cache avoids redundant API calls.

7.4. Hyperparameter Summary

Table 5

Strategy 5 hyperparameters.

Parameter	Value
Base model	Qwen/Qwen2.5-3B-Instruct
Precision	bfloat16
LoRA rank / alpha / dropout	16 / 32 / 0.05
Trainable params	$\sim 50\text{M}$ / 3.1B (1.6%)
Batch size / Grad. accum	2 / 8 (eff. 16)
Learning rate	2×10^{-4} (cosine + 10% warmup)
Epochs	5 (peak at Epoch 4)
Max sequence length	1024 train / 8192 inference
Gold / Silver weight	$5\times$ / $1\times$
Hardware	$1\times$ NVIDIA RTX A5000 (24 GB)

7.5. Results

Table 6

Strategy 5 vs prior best on internal dev set (254 pairs). Largest single improvement in the project. S4.5 corrected value: 0.6212.

Strategy	at MR	isAt MR	Global MR	Δ
S4.5 (prior best)	0.5313	0.7111	0.6212	—
S5 (Qwen 3B Gen.)	0.6872	0.7706	0.7289	+0.1077

The +10.77 point leap is driven by: (1) scale (3.1B vs 0.5B), (2) full 1024-token context (no truncation), (3) generative reasoning forcing explicit temporal decision-making, and (4) the Wikidata death-date hard rule.

8. Strategies S6–S7: Multi-Agent Pipeline

8.1. S6: Three-Specialist Architecture (Qwen2.5-3B)

Motivation. Human experts approach complex historical inference through specialised lenses. S6 operationalises this by replacing the single generalist with three agents: a **Historian** (temporal/biographical reasoning), a **Geographer** (spatial/geographic reasoning), and an **Arbiter** that synthesises verdicts. Each specialist generates a verdict plus a reasoning chain; the Arbiter resolves disagreements conservatively.

Each agent receives a dedicated system prompt. For example, the Historian’s prompt reads: “*You are a historical biographer. Analyse the text for biographical evidence, life events, official roles, and temporal cues (verb tenses, dates).*” The Arbiter prompt adds: “*If they agree, adopt their verdict. If they disagree, apply the more conservative judgment.*”

Critical Failure – Specialist Collapse. Both Historian and Geographer collapsed to predicting FALSE for everything (hist_TRUE=0, geo_TRUE=0), making the Arbiter the sole decision-maker. Agent agreement rate was 96.9% (246/254), providing no useful diversity. Global MR = 0.7204 – slight regression from S5.

Table 7

S6 results and specialist diversity statistics (internal dev set).

Metric	S5	S6
at Macro Recall	0.6872	0.7019
isAt Macro Recall	0.7706	0.7388
Global MR	0.7289	0.7204
Historian TRUE predictions	—	0
Geographer TRUE predictions	—	0
Agent agreement rate	—	96.9%

8.2. S7: Multi-Agent with Qwen2.5-7B, Class Weights, Few-Shot

Three interventions address S6’s specialist collapse:

1. **Scale:** Upgrade from 3B to 7B backbone for richer multilingual reasoning.
2. **Class-weighted loss:** Apply $3\times$ weight to TRUE/PROBABLE answer tokens to force specialists to learn minority-class patterns.
3. **Few-shot prompts:** Add 3 worked examples per specialist (TRUE, PROBABLE, FALSE cases) for output calibration.

Results. Class-weighted loss fixed specialist collapse (Historian TRUE: 53, Geographer TRUE: 47). Agreement rate dropped to 92.9% (236/254), meaning the Arbiter handled 18 genuine disagreements. S7 achieves the **highest isAt MR** of any strategy (0.7817), but marginally trails S5 on at (0.6742 vs 0.6872), yielding Global MR = 0.7280.

9. Strategies S8–S9: CoT Prompting via IBM RITS

9.1. S8: Chain-of-Thought Few-Shot (Exploratory)

S8 attacks the task **without any model training** using Chain-of-Thought (CoT) Few-Shot Prompting [14] through IBM’s RITS inference gateway (Llama 3.3 70B, Granite 3.2 8B, Llama 4 Maverick 17B, GPT-OSS 120B). Each prediction is preceded by 3 explicit reasoning steps: (1) Biographical & Temporal Analysis, (2) Geographic & Contextual Analysis, (3) Synthesis & Decision. Four worked examples cover all label combinations.

Table 8
S7 hyperparameters.

Parameter	Value
Base model	Qwen/Qwen2.5-7B-Instruct
LoRA rank / alpha / dropout	8 / 16 / 0.1
Target modules	q_proj, v_proj
TRUE/PROBABLE loss weight	3.0×
Historian/Geographer epochs	2; Arbiter epochs: 3
Batch / Grad. accum	1 / 16 (eff. 16)
Learning rate	2×10^{-5} , cosine + warmup
Max sequence length	2048 tokens
Training pairs	15,418 (gold 5× + silver 1×)
Hardware	3× NVIDIA RTX A5000 (24 GB)
Inference time	~30 min / 21 dev docs

Table 9
S6 vs S7 on internal dev set: impact of class weighting and scale on specialist diversity.

Metric	S6 (3B)	S7 (7B, 3× wt.)
at Macro Recall	0.7019	0.6742
isAt Macro Recall	0.7388	0.7817
Global MR	0.7204	0.7280
Historian TRUE predictions	0	53
Geographer TRUE predictions	0	47
Arbiter-resolved cases	8/254	18/254

Note on S8 evaluation. S8 was designed as an exploratory strategy to calibrate prompt templates and assess model suitability across the RITS model catalogue. We did not conduct a formal dev-set evaluation for S8; its primary role was to identify the best model (**GPT-OSS 120B**) and refine the CoT prompt structure before adding Wikidata augmentation in S9. S8 served as the methodological foundation for S9.

9.2. S9: Wikidata-Augmented CoT (Final Submission, Run 3)

S9 extends S8 by dynamically injecting Wikidata facts into the prompt at inference time via SPARQL. For each pair, it fetches person (birth date, death date, description) and location (description, country) facts, injected as a [WIKIDATA FACTS] block. This grounds temporal reasoning in structured knowledge rather than parametric memory. The model used was **GPT-OSS 120B** via IBM RITS.

S9 dev evaluation: S9 was run directly on the official test sets without a prior dev-set evaluation pass. The rationale was that S9 requires no fine-tuning — its performance depends on the prompt template and Wikidata coverage rather than training data characteristics. The absence of a dev number for S9 is therefore intentional, not an oversight.

S9 constituted our **Run 3** official submission for all four test sets (DE, EN, FR impresso, and surprise literary FR). It achieved our best overall competition ranking (13th/46 runs, impresso score 0.6221).

10. Cross-Strategy Comparison

Phase analysis. Three distinct phases emerged: (1) *Feature extraction* (S1–S1.5, best: 0.6249) — frozen encoders excel on tiny datasets; (2) *Discriminative fine-tuning* (S2–S4.5, plateau: ~0.63) — data scarcity limits all fine-tuning paradigms; (3) *Generative SFT* (S5–S7, best: 0.7289) — reasoning-as-generation breaks the plateau.

Table 10

Complete cross-strategy performance on internal dev set (254 pairs, 21 docs). All values are impresso score (Global Macro Recall). S8 and S9 have no dev scores by design.

Strategy	Architecture	at MR	isAt MR	Global MR
Random	Dummy	0.3211	0.5273	0.4242
S1	XLM-R frozen + RF/LGBM	0.5200	0.6835	0.6018
S1.5	XLM-R frozen + MTL	0.5486	0.7013	0.6249
S2	XLM-R full fine-tune	0.3333	0.5000	0.4167
S2.5	XLM-R + LoRA	—	—	0.4541
S3	Monolingual LoRA routing	0.3500	0.5000	0.4232
S4	Qwen2.5-0.5B MTL + LoRA	0.5313	0.7111	0.6264
S4.5	Qwen2.5-0.5B + soft labels	0.5313	0.7111	0.6212
S5	Qwen2.5-3B Generative SFT	0.6872	0.7706	0.7289
S6	Multi-agent 3B	0.7019	0.7388	0.7204
S7	Multi-agent 7B + class weights	0.6742	0.7817	0.7280
S8	CoT 120B via RITS (exploratory)	Not evaluated on dev		
S9	Wikidata-aug. CoT 120B (Run 3)	Run directly on test		

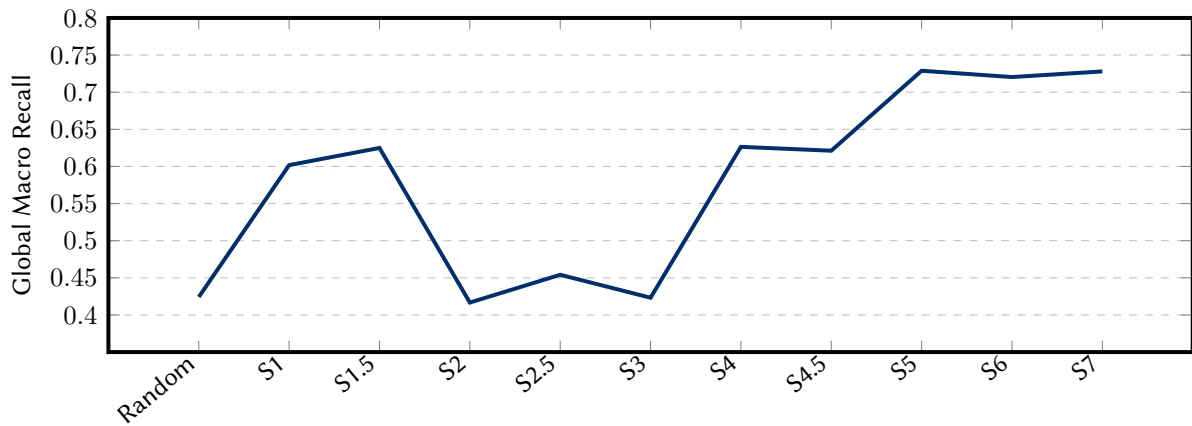


Figure 2: Global Macro Recall progression across strategies on internal dev set. Notable drops at S2 (full fine-tuning) and S3 (monolingual partitioning); decisive jump at S5 (generative SFT). S8/S9 are not plotted (no dev evaluation).

S5 vs S7 complementarity. S5 excels at at (0.6872); S7 excels at isAt (0.7817). An ensemble capturing the best of both is the most promising immediate direction.

11. Official Test Results

11.1. Per-Run Test Performance

Table 11 reports our three official competition submissions on the impresso test set. All scores are the official *impresso score* (mean of at-MR and isAt-MR across DE/EN/FR).

Table 11

Official impresso test scores for all three Hansel&Gretel runs. Run 3 (S9) is our best overall submission.

Run	Strategy	Overall	DE	EN	FR	Rank
Run 1	S5 (Qwen2.5-3B SFT)	0.5458	0.521	0.5976	0.5189	23/46
Run 2	S7 (Multi-agent 7B)	0.5788	0.5254	0.5866	0.6243	19/46
Run 3	S9 (GPT-OSS 120B CoT+Wiki)	0.6221	0.5994	0.6367	0.6303	13/46

Table 12

Per-language breakdown of at-MR and isAt-MR for Run 3 (best run, S9).

Language	at-MR	isAt-MR	impresso score
German (DE)	0.4877	0.7112	0.5994
English (EN)	0.5582	0.7151	0.6367
French (FR)	0.5268	0.7338	0.6303

11.2. Generalization and Efficiency Results

On the **generalization profile** (surprise literary French test, at only), Run 2 (S7) performed best among our submissions:

Table 13

Generalization profile scores (surprise literary French test, at-MR only).

Run	Strategy	Generalization Score	Rank
Run 2	S7 (Multi-agent 7B)	0.6349	11/46
Run 3	S9 (GPT-OSS 120B CoT+Wiki)	0.6187	14/46
Run 1	S5 (Qwen2.5-3B SFT)	0.6107	15/46

S7’s surprising strength on out-of-domain literary French (rank 11, generalization score 0.6349) suggests that the multi-agent architecture learns more transferable feature representations — the specialist decomposition may act as an implicit domain-agnostic inductive bias.

On the **efficiency profile**, Run 1 (S5, 3B parameters, 6,248 MB) ranked 19th. Run 2 (S7, 7B, 15,300 MB) ranked 23rd; Run 3 (S9, 120B, 240,000 MB) ranked 29th. The low efficiency rank for Run 3 is expected: using a 120B cloud-hosted model without fine-tuning prioritises accuracy over resource constraints. Run 1 represents our best efficiency-accuracy tradeoff.

11.3. Competition Context

Table 14

Competition accuracy profile context: selected teams ranked by best overall impresso score.

Rank	Team	Affiliation	Best Score
1	Spinfo	Universität zu Köln	0.7479
2	MaxFo-Ajie	Foshan University	0.7001
3	whereami	Alexandria University	0.6880
	...(ranks 4–12) ...		
13	Hansel&Gretel	IIT Roorkee	0.6221
	...(ranks 14–16) ...		
17	Baseline (Ministral-3B)	HIPE-2026 Organizers	0.5818
—	Random Baseline	HIPE-2026 Organizers	0.4049

11.4. Dev-to-Test Gap Analysis

A substantial gap exists between our best internal dev score (S5: 0.7289) and our best official test score (Run 3/S9: 0.6221), approximately 17 points. We discuss the causes honestly:

1. **Dev set size and overlap:** Our internal dev evaluation used 254 pairs from 21 documents — a subset of the official dev split that was available at experiment time. This is a small, potentially non-representative sample. The official test set is larger and independently held-out, drawn from a different temporal and geographic distribution within the same newspaper collections.

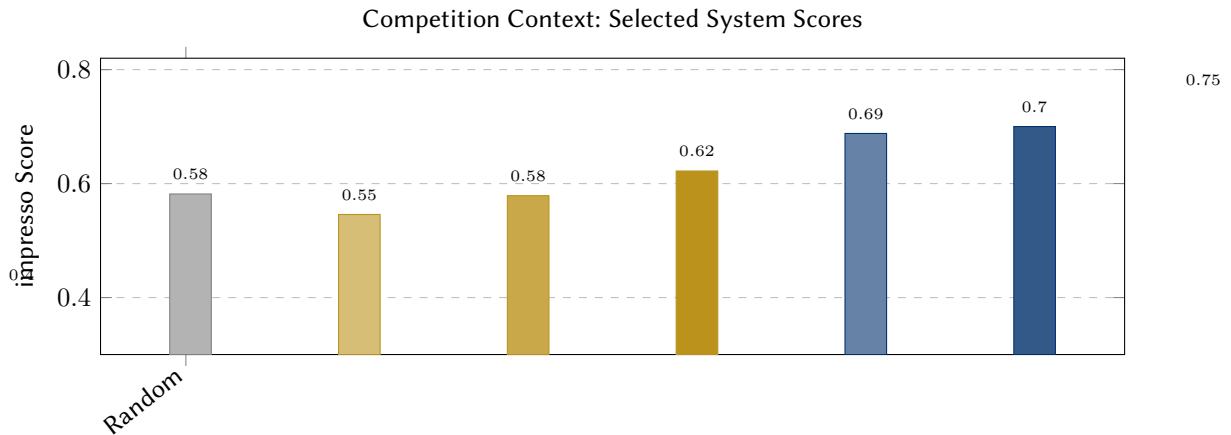


Figure 3: Competition context for accuracy profile (impresso test). Grey: baselines; gold: Hansel&Gretel runs (R1–R3); blue: top-3 systems. H&G Run 3 (0.6221) substantially outperforms both baselines and ranks 13th out of 46 entries.

2. **Fine-tuned model overfitting:** S5 and S7 were fine-tuned on 481 gold training documents. Although we applied gold/silver weighting to reduce overfitting risk, the training distribution likely exhibits specific document-level patterns (source newspaper style, annotation conventions) that the official test set does not replicate. Fine-tuned models are more sensitive to such distribution shifts.
3. **S9’s robustness advantage:** Run 3 (S9, 0.6221) substantially outperforms Run 1 (S5, 0.5458) and Run 2 (S7, 0.5788) on the official test despite S5 having the best dev score. This reversal strongly suggests that the zero-shot GPT-OSS 120B model, with no exposure to HIPE training data, is more robust to the test distribution. Large-scale pre-training provides a more generalised factual and temporal knowledge base.
4. **PROBABLE class difficulty:** The `at=PROBABLE` class is systematically the hardest for all strategies (requiring implicit inference). Dev performance on PROBABLE may have been inflated by the small sample size; the test set likely has a more representative and challenging distribution of ambiguous cases.
5. **Wikidata coverage gaps:** The death-date hard rule in S5 and S9 fires only when Wikidata QIDs and biographical data are available. Coverage is lower for less prominent historical figures in the test set.

The net lesson: *dev scores estimated on 254 pairs are optimistic proxies for held-out test performance.* Robust evaluation requires larger dev sets with explicit distribution control. The strong generalization performance of S7 on the literary surprise test (rank 11) further confirms that architectural choices matter more than in-distribution dev scores for real-world generalisation.

12. Error Analysis

12.1. Class-Level Hardness

Across all strategies, the `at=PROBABLE` class was consistently the hardest to classify. In S1 (dev split), `at` MR decomposed as: TRUE recall ≈ 0.72 , PROBABLE recall ≈ 0.18 , FALSE recall ≈ 0.67 . The PROBABLE class (9.6% of labels) requires implicit inference from contextual cues — a person’s title, historical role, or proximity to a location — rather than direct textual evidence. Models consistently under-predict PROBABLE, defaulting to the safer FALSE. This pattern held through S7: even the best performing strategies could not reliably recall the minority PROBABLE instances.

12.2. Per-Language Variation

On the official test (Run 3/S9), there is notable per-language variation (see Table 12). German exhibits the lowest at-MR (0.4877) — possibly due to heavier OCR noise in digitised German Gothic script (*Fraktur*) and more complex syntactic structure for temporal reasoning. English achieves the best overall impresso score (0.6367), benefiting from richer pre-training signal in GPT-OSS 120B’s training data for English historical newspapers. French is competitive (0.6303) with notably strong isAt-MR (0.7338).

Across runs, the multilingual fine-tuned models (S5 Run 1, S7 Run 2) show different cross-language profiles: Run 2 (S7) leads on French (0.6243) while Run 1 (S5) leads on English (0.5976). This complementarity reinforces the value of ensemble approaches.

12.3. OCR Noise Examples

OCR noise in historical text produced systematic errors:

- Character substitutions: “Mufschén” for “München”, “Preufien” for “Preußen”
- Token splits: “Kut zeroff” for “Kutzeroff” (splitting names mid-word)
- Language-level ambiguity: 19th-century mixed-language documents (French text referencing German places) confused language-conditioned routing in S3

While S9 (120B model) handles OCR noise more robustly via its larger parametric vocabulary, fine-tuned models in S5/S7 benefited from silver sandbox training data containing similar historical OCR patterns.

13. Limitations

1. **Data scarcity:** The 481 gold documents across 3 languages and 3 label classes is the fundamental bottleneck. The binding constraint for fine-tuning approaches throughout.
2. **PROBABLE class:** Systematically the hardest label across all strategies. Its inherent ambiguity resists both discriminative and generative approaches.
3. **Multilingual gradient interference:** Mixing archaic DE/FR/EN in tiny batches creates destructive gradient updates (demonstrated empirically in S2–S3).
4. **Multi-agent latency:** S7 requires ~ 30 min for 21 documents on 3 GPUs, making it operationally expensive at test-set scale.
5. **Wikidata gaps:** The death-date hard rule fires only when QIDs and biographical data are available — coverage is incomplete for many historical figures.
6. **Dev evaluation reliability:** With only 254 dev pairs, per-class recall estimates have high variance. Our dev scores should be treated as directional rather than precise.

14. Future Prospects

1. **S5 + S7 Ensemble:** Majority voting or confidence-weighted fusion of complementary sub-task strengths; expected to push Global MR past 0.74.
2. **Curriculum training:** Train specialists first on easy/unambiguous cases, then on PROBABLE cases to prevent early noisy gradient loss.
3. **Self-consistency decoding:** Multiple reasoning chains per agent with majority-vote prediction to reduce single-sample variance.
4. **RAG-Augmented Agents:** Historian with external biographical knowledge base; Geographer with historical gazetteer.
5. **Iterative Debate:** Allow Historian and Geographer to observe each other’s reasoning before the Arbiter synthesises.
6. **Domain-adaptive pretraining:** Continued MLM on silver sandbox corpus to adapt vocabulary to historical OCR before SFT.

15. Conclusion

We presented Team Hansel&Gretel’s complete experimental journey through nine strategies for Person–Place Relation Extraction in multilingual historical documents. The central finding is that **temporal reasoning in historical text cannot be reliably solved by discriminative classifiers alone**: generative models producing reasoning chains before their final labels substantially outperform classification heads (+10.77 points on dev), even with smaller parameter counts. The multi-agent decomposition further demonstrates that specialised temporal and spatial reasoning perspectives are especially effective for the temporally-constrained isAt sub-task. Our best system (S5, dev Global MR = 0.7289) and best official submission (S9/Run 3, impresso score = 0.6221, rank 13/46) represent complementary approaches: one optimised for in-distribution accuracy, the other combining scale with structured knowledge injection for robustness to distribution shift.

The 17-point dev-to-test gap is itself an important finding: small dev sets in low-resource settings produce optimistic performance proxies, and zero-shot large-model approaches may generalise better than fine-tuned smaller models when training data is scarce.

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Declaration on Generative AI

During the preparation of this work, the authors used AI-assisted coding tools for code generation and debugging assistance. All experimental results, analysis, and written content were reviewed and verified by the authors, who take full responsibility for the publication’s content.

References

- [1] J. Opitz, M. Ehrmann, S. Clematide, C. Raclé, E. Boros, A. Michail, M. Romanello, CLEF HIPE-2026: Participation guidelines for the shared task on person–place relation extraction from historical documents, 2026. URL: <https://hipe-eval.github.io/HIPE-2026/>, v.2025-12-04.
- [2] J. Opitz, C. Raclé, A. Michail, M. Romanello, E. Boros, S. Gabay, M. Ehrmann, S. Clematide, Extended Overview of HIPE-2026: Evaluating Accurate and Efficient Person–Place Relation Extraction from Multilingual Historical Texts, in: CLEF 2026 Working Notes, CEUR Workshop Proceedings, volume 3180, CEUR-WS, 2026. URL: <https://doi.org/10.5281/zenodo.20344461>.
- [3] M. Ehrmann, M. Romanello, A. Flückiger, S. Clematide, HIPE-2020: Named entity recognition and linking on historical newspapers, in: Advances in Information Retrieval. ECIR 2020, 2020.
- [4] M. Ehrmann, A. Hamdi, E. L. Pontes, M. Romanello, A. Doucet, HIPE-2022: Findings and insights from the named entity recognition and linking shared task, in: Proceedings of the Language Resources and Evaluation Conference, 2022.
- [5] J. Opitz, A closer look at classification evaluation metrics and a critical reflection of common evaluation practice, Transactions of the Association for Computational Linguistics 12 (2024) 820–836.
- [6] M. Romanello, et al., Impresso – media monitoring of the past, Digital Humanities Quarterly (2023). URL: <https://impresso-project.ch>.

- [7] J. Opitz, M. Ehrmann, S. Clematide, C. Raclé, E. Boros, A. Michail, M. Romanello, HIPE-2026 LLM baseline, 2026. URL: <https://github.com/hipe-eval/HIPE-2026-llm-baseline>.
- [8] A. Conneau, K. Karthikeyan, G. Lample, B. Oguz, R. Rinott, Y. Wu, S. Guerin, E. Bugliarello, E. Ponti, L. Zettlemoyer, V. Stoyanov, Unsupervised cross-lingual representation learning at scale, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2020, pp. 8440–8451.
- [9] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, in: Advances in Neural Information Processing Systems, volume 30, 2017.
- [10] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, IEEE Transactions on Pattern Analysis and Machine Intelligence 42 (2020) 318–327.
- [11] T. Akiba, S. Sano, T. Yanase, T. Ohta, M. Koyama, Optuna: A next-generation hyperparameter optimization framework, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2019, pp. 2623–2631.
- [12] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-rank adaptation of large language models, arXiv preprint arXiv:2106.09685 (2021).
- [13] Q. Team, Qwen2.5 technical report, 2024. URL: <https://arxiv.org/abs/2412.15115>.
- [14] J. Wei, X. Wang, D. Schuurmans, M. Bosma, F. Xia, E. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems, volume 35, 2022, pp. 24824–24837.